# Inversion and projective methods

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L5-GE1 Level 5 Advanced · Geometry

Inversion, cross-ratio, harmonic division, poles and polars, Brocard's theorem and projective maps, with complete proofs.
**By the end you can** choose a centre and radius of inversion and translate a problem through it; recognise and prove harmonic ranges, and use poles, polars and Brocard's theorem; use a projective map to send a line to infinity and write the argument rigorously.
**Prerequisites:** Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

## 1 Lesson

*Prove that for any four points $A, B, C, D$ in the plane, $AB \cdot CD + AD \cdot BC \geq AC \cdot BD$.* (Ptolemy's inequality.) There is no circle in the statement and the terms are products of lengths. Invert about $A$ with radius 1 (defined below). Theorem 1.1 gives $B'C' = BC/(AB \cdot AC)$, and similarly for the pairs $C, D$ and $B, D$, so the triangle inequality $B'C' + C'D' \geq B'D'$ reads

$$\frac{BC}{AB \cdot AC} + \frac{CD}{AC \cdot AD} \geq \frac{BD}{AB \cdot AD}.$$

Multiplying by $AB \cdot AC \cdot AD$ gives the result. Equality holds if and only if $C'$ lies on segment $B'D'$, and by Theorem 1.2 this happens if and only if $A, B, C, D$ lie on a circle (or a line) with $C$ on the arc $BD$ not containing $A$.

This unit covers inversion, which turns circles through a chosen point into lines, then cross-ratio, harmonic division and polars, and finally projective maps, which send a chosen line to infinity. Power of a point, radical axes and directed angles (L4-GE1, L4-GE2) are assumed.

### 1.1 Inversion

Let $\omega$ be a circle with centre $O$ and radius $r$. The *inverse* of a point $P \neq O$ is the point $P'$ on ray $OP$ with $OP \cdot OP' = r^2$. Inversion is its own inverse, fixes $\omega$ pointwise and swaps its inside and outside. We add a point at infinity and say that inversion swaps $O$ and $\infty$.

**Theorem 1.1.**
If $P, Q \neq O$ and $O, P, Q$ are not collinear, then triangles $OPQ$ and $OQ'P'$ are similar, and

$$P'Q' = \frac{r^2 \cdot PQ}{OP \cdot OQ}.$$

*Proof.* From $OP \cdot OP' = OQ \cdot OQ'$ we get $OP/OQ = OQ'/OP'$, and the angle at $O$ is common, so $OPQ \sim OQ'P'$ with ratio $OQ'/OP = r^2/(OP \cdot OQ)$. The collinear case is a direct computation. ■

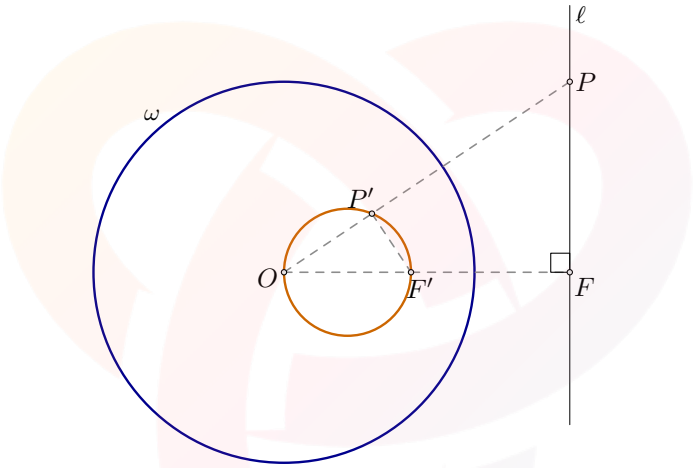


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**Theorem 1.2.**
(1) A line through $O$ is mapped to itself. (2) A line not through $O$ is mapped to a circle through $O$, and conversely. (3) A circle not through $O$ is mapped to a circle not through $O$.

*Proof.* Statement (1) is in the definition. For (2), let $F$ be the foot of the perpendicular from $O$ to the line $\ell$. For $P$ on $\ell$, Theorem 1.1 gives $\angle OP'F' = \angle OFP = 90°$, so $P'$ lies on the circle with diameter $OF'$; running the argument backwards shows every point of that circle other than $O$ is the image of a point of $\ell$. Its tangent at $O$ is parallel to $\ell$. For (3), let $\gamma$ be the circle and $k \neq 0$ the power of $O$ with respect to $\gamma$. For $P$ on $\gamma$, line $OP$ meets $\gamma$ again at $Q$ with $\overrightarrow{OP} \cdot \overrightarrow{OQ} = k$ (with $Q = P$ for a tangent). Then $\overrightarrow{OP'} = \frac{r^2}{OP^2}\overrightarrow{OP} = \frac{r^2}{k}\overrightarrow{OQ}$, so $P'$ is the image of $Q$ under the homothety with centre $O$ and ratio $r^2/k$. Hence $\gamma'$ is the image of $\gamma$ under that homothety. ■

The proof of (3) shows that the centre of $\gamma$ is *not* mapped to the centre of $\gamma'$; the two centres are only collinear with $O$.




**Theorem 1.3.**
Inversion preserves the angle between two curves (lines or circles) at a common point $P \neq O$. In particular tangent curves are mapped to tangent curves, and orthogonal circles to orthogonal circles.

*Proof.* First, two different circles or lines touch at $P$ if and only if $P$ is their only common point, and inversion is a bijection, so curves touching at $P \neq O$ go to curves touching at $P'$. Now let $\gamma$ pass through $P$ with tangent $t$ at $P$, and let $\delta$ be the circle through $O$ tangent to $t$ at $P$ (the line $OP$ if $t$ passes through $O$). Then $\delta$ touches $\gamma$ at $P$ (or equals it), so the line $\delta'$ is the tangent to $\gamma'$ at $P'$, and by the note in the proof of 1.2(2), $\delta'$ is parallel to the tangent to $\delta$ at $O$. The tangents to $\delta$ at $O$ and at $P$ are mirror images in the perpendicular bisector of $OP$, so the tangent to $\gamma'$ at $P'$ makes with line $OP$ the angle opposite to the one $t$ makes. Doing this for two curves through $P$, the angle between them keeps its size and reverses orientation. ■

## 1.2 Choosing the centre and the radius

Rules for choosing, in order of how often they apply:

1. Invert at a point through which several circles pass; they become lines. A point where two circles touch is the best choice, as the two circles become parallel lines.

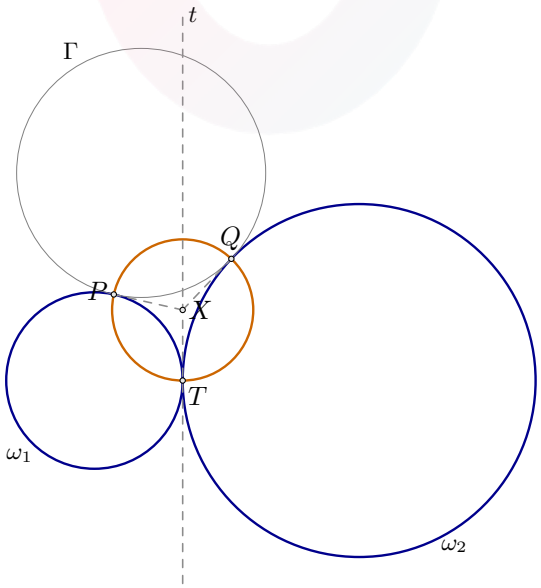


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2. If a product such as $AX \cdot AY$ is fixed, invert at $A$ with radius $\sqrt{AX \cdot AY}$, so that $X$ and $Y$ swap.
3. ($\sqrt{bc}$ inversion.) In triangle $ABC$, invert at $A$ with radius $\sqrt{AB \cdot AC}$ and then reflect in the bisector of angle $A$. This map swaps $B$ and $C$, and swaps line $BC$ with the circumcircle $\Gamma$. For example, the foot $D$ of the altitude goes to the point of $\Gamma$ on the reflection of line $AD$, which is line $AO$; so it goes to the antipode $A^*$ of $A$, and $AD \cdot AA^* = AB \cdot AC$, the formula $2R \cdot AD = bc$.
4. Before solving, write down what every object becomes.

**Example 1.**
Circles $\omega_1$ and $\omega_2$ touch externally at $T$. A circle $\Gamma$, not through $T$, touches $\omega_1$ at $P$ and $\omega_2$ at $Q$, and $T, P, Q$ are not collinear. Prove that the circle through $T, P, Q$ meets $\omega_1$ and $\omega_2$ at right angles, and that its centre lies on the common tangent of $\omega_1$ and $\omega_2$ at $T$.

*Solution.*
Three of the four circles pass through $T$, so we invert at $T$ with any radius. The images $\ell_1, \ell_2$ of $\omega_1, \omega_2$ are lines, and they have no common point (the only common point of $\omega_1, \omega_2$ is $T$, which goes to $\infty$), so they are parallel. Each is parallel to the common tangent $t$ at $T$, by the note in Theorem 1.2. The circle $\Gamma$ becomes a circle $\Gamma'$ touching $\ell_1$ at $P'$ and $\ell_2$ at $Q'$ (Theorem 1.3), so $P'Q'$ is a diameter of $\Gamma'$ perpendicular to both lines.
The circle $TPQ$ passes through $T$, so its image is the line $P'Q'$. This line does not pass through $T$, since otherwise $T, P', Q'$ and hence $T, P, Q$ would be collinear. The line $P'Q'$ meets $\ell_1$ at right angles, so by Theorem 1.3 the circle $TPQ$ meets $\omega_1$ at right angles at $P$; similarly for $\omega_2$ at $Q$.
For the centre $X$ of circle $TPQ$: orthogonality at $P$ means $XP$ is tangent to $\omega_1$, so the power of $X$ with respect to $\omega_1$ is $XP^2$. Similarly its power with respect to $\omega_2$ is $XQ^2 = XP^2$. Hence $X$ lies on the radical axis of $\omega_1$ and $\omega_2$, which is $t$.




## 1.3 Cross-ratio and harmonic division

For collinear points $A, B, C, D$ define


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$$(A,B;C,D) = \frac{CA}{CB} \div \frac{DA}{DB}$$

with signed lengths. If four lines through a point $P$ meet a line at $A, B, C, D$, the area formula gives $\frac{CA}{CB} = \frac{PA \sin \angle CPA}{PB \sin \angle CPB}$, and the same for $D$, so $PA$ and $PB$ cancel and $(A,B;C,D)$ depends only on the four lines (using directed angles for signs). We write $P(A,B;C,D)$ for this common value: **cross-ratio is preserved by projecting from a point onto another line.** For four points on a circle, $P(A,B;C,D)$ is the same for every $P$ on the circle, by inscribed angles, and we call it $(A,B;C,D)$ on the circle.

We call $A,B;C,D$ *harmonic* if $(A,B;C,D) = -1$, that is, $C$ and $D$ divide $AB$ internally and externally in the same ratio. Note that $(A,B;C,\infty) = CA/CB$, so $(A,B;C,\infty) = -1$ if and only if $C$ is the midpoint of $AB$.

**Theorem 3.1.**
(1) If $C, D$ on line $AB$ are the feet of the internal and external bisectors from $P$ in triangle $PAB$, then $(A,B;C,D) = -1$. Conversely, if $(A,B;C,D) = -1$ and $\angle CPD = 90^\circ$, then $PC$ and $PD$ bisect the angles between $PA$ and $PB$.
(2) Let cevians $AD, BE, CF$ of triangle $ABC$ be concurrent, and let line $EF$ meet line $BC$ at $T$. Then $(B,C;D,T) = -1$.
(3) If $(A,B;C,D) = -1$ and $M$ is the midpoint of $AB$, then $MC \cdot MD = MA^2$.
(4) If points $X, Y, Z$ on a line have coordinates $x, y, z$ measured from a point $P$ of the line, then $(X,Y;P,Z) = -1$ if and only if $\frac{2}{z} = \frac{1}{x} + \frac{1}{y}$.

Statement (1) is the angle bisector theorem: $\frac{CA}{CB} = \frac{PA}{PB} = \frac{DA}{DB}$ in absolute value, with opposite signs. For the converse, the circle with diameter $CD$ passes through $P$ and is the Apollonius circle $\{X : XA/XB = CA/CB\}$, so $PA/PB = CA/CB$ and $PC$ is the internal bisector. For (2), Ceva gives $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ and Menelaus for the line $EFT$ gives $\frac{BT}{TC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$ (signed); dividing, $\frac{BD}{DC} = -\frac{BT}{TC}$. Statements (3) and (4) are short computations: for (3) put $M = 0$, $A = -1$, $B = 1$, and $(A,B;C,D) = -1$ becomes $cd = 1$; for (4), $(X,Y;P,Z) = \frac{x}{y} \cdot \frac{z-y}{z-x} = -1$ rearranges to $z(x+y) = 2xy$.

Statement (2) is usually applied to a quadrilateral. If $ABCD$ is a quadrilateral with $Q = AB \cap CD$, $R = AD \cap BC$, $P = AC \cap BD$, then in triangle $QAD$ the cevians $AC$, $DB$ and $QP$ concur, so if $QP$ meets $AD$ at $X$, then $(A,D;X,R) = -1$. You should be able to see this harmonic range instantly.

## 1.4 Poles and polars

The *polar* of a point $P \neq O$ with respect to $\omega$ is the line through the inverse $P'$ perpendicular to $OP$; $P$ is the *pole* of that line. If $P$ is outside $\omega$ and $PT$ is a tangent, then $OT^2 = OP \cdot OP'$ shows that $P'$ is the foot of $T$ on $OP$, so the polar of $P$ is the chord joining the two points of contact. The polar of a point of $\omega$ is the tangent there. With $O$ as origin, the polar of $P$ is the line $\{X : X \cdot P = r^2\}$ (dot product).

**Theorem 4.1 (La Hire).**
$Q$ lies on the polar of $P$ if and only if $P$ lies on the polar of $Q$.
**Theorem 4.2.** If a line through $P$ (not on $\omega$) meets $\omega$ at $X, Y$ and the polar of $P$ at $Z$, then $(X,Y;P,Z) = -1$.
**Theorem 4.3 (Brocard).** Let $ABCD$ be inscribed in $\omega$, and suppose $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$ exist. Then $QR$, $RP$, $PQ$ are the polars of $P$, $Q$, $R$. Hence $O$ is

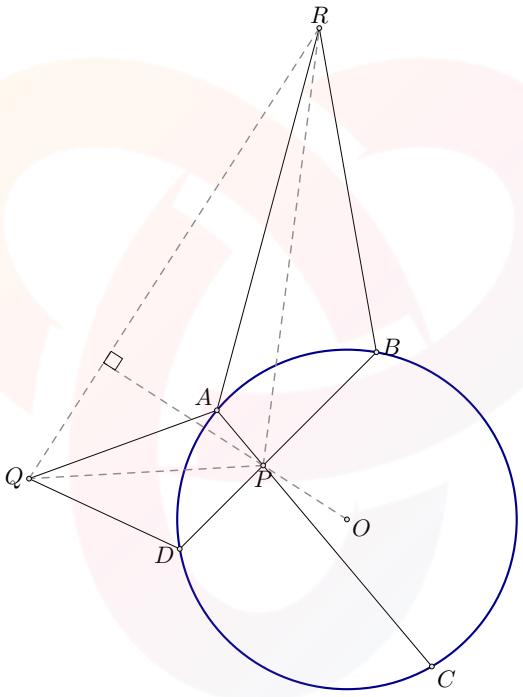


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the orthocentre of triangle $PQR$.

Theorem 4.1 is trivial in coordinates: both statements say $P \cdot Q = r^2$. For 4.2, take $O$ as origin, $r = 1$, and write the line as $P + tu$ with $|u| = 1$. The parameters $t_1, t_2$ of $X, Y$ are the roots of $t^2 + 2(P \cdot u)t + |P|^2 - 1 = 0$, and $Z$ has parameter $t_3 = (1 - |P|^2)/(P \cdot u)$. Then $\frac{1}{t_1} + \frac{1}{t_2} = \frac{-2P \cdot u}{|P|^2 - 1} = \frac{2}{t_3}$, which is Theorem 3.1(4). If $P \cdot u = 0$, then $Z = \infty$ and $t_2 = -t_1$, so again $(X, Y; P, Z) = -1$.

*Proof of Theorem 4.3.* Let $QP$ meet $AD$ at $X$ and $BC$ at $Y$. By the quadrilateral form of Theorem 3.1(2), $(A, D; X, R) = -1$, and in the same way (triangle $QBC$) $(B, C; Y, R) = -1$. By Theorem 4.2 and the uniqueness of the harmonic conjugate, $X$ and $Y$ lie on the polar of $R$, so that polar is line $XY = PQ$. Similarly the polar of $Q$ is $PR$. Now $P$ lies on the polars of $Q$ and $R$, so by La Hire, $Q$ and $R$ lie on the polar of $P$, which is therefore $QR$. Finally a polar is perpendicular to the line joining its pole to $O$, so $OP \perp QR$, $OQ \perp RP$, $OR \perp PQ$. ■




**Example 2.**
The incircle of triangle $ABC$, with centre $I$ and $AB \neq AC$, touches $BC, CA, AB$ at $D, E, F$. Line $EF$ meets line $BC$ at $T$. Prove that $(B, C; D, T) = -1$ and $IT \perp AD$.

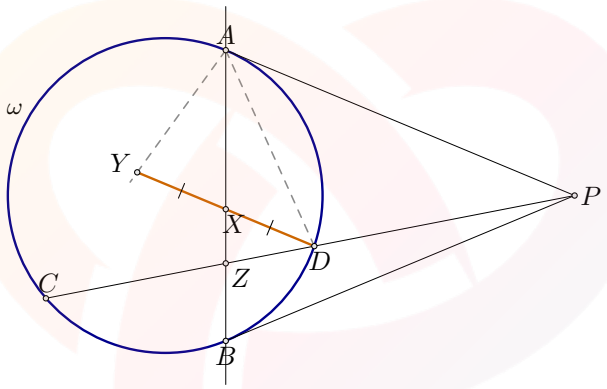
*Solution.*
The point $T$ exists: $EF \perp AI$, and $BC \perp AI$ only if $AB = AC$. Since $BD = BF$, $CD = CE$ and $AE = AF$, Ceva shows that $AD, BE, CF$ are concurrent, and Theorem 3.1(2) gives $(B, C; D, T) = -1$.
For the perpendicularity, we should instantly spot $A$ as the pole of $EF$ (it is the chord of contact from $A$). Since $T$ lies on the polar of $A$, La Hire says $A$ lies on the polar of $T$. Since $T$ lies on $BC$, which is the polar of $D$ (the tangent at $D$), $D$ lies on the polar of $T$. Hence the polar of $T$ is $AD$, and it is perpendicular to $IT$.



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**Example 3.**
From a point $P$ outside a circle $\omega$, tangents touch $\omega$ at $A$ and $B$, and a line through $P$ meets $\omega$ at $C$ and $D$. The line through $D$ parallel to $PA$ meets line $AB$ at $X$ and line $AC$ at $Y$. Prove that $X$ is the midpoint of $DY$.

*Solution.*
A midpoint and a parallel line suggest a harmonic range containing a point at infinity. Let line $PCD$ meet $AB$ at $Z$; it does meet it, because the line through $P$ parallel to $AB$ is perpendicular to $OP$ at $P$ and misses $\omega$. Since $AB$ is the polar of $P$, Theorem 4.2 gives $(C, D; P, Z) = -1$.
Now project from $A$ onto the line $m$ through $D$ parallel to $AP$. The lines $AC, AD, AP, AZ$ meet $m$ at $Y$, $D$, the point at infinity, and $X$ respectively. Hence $(Y, D; \infty, X) = -1$, so $XY = -XD$ as signed lengths and $X$ is the midpoint of $DY$. Note that $C, D \neq A$, since the secant is not the tangent $PA$; so $A$ is not on $m$, and the lines $AC$ and $AB$ are not parallel to $m$, which means $X$ and $Y$ exist.




**Example 4.**
Let $ABCD$ be a convex cyclic quadrilateral with no two sides parallel. Lines $AB$ and $CD$ meet at $Q$, and lines $AD$ and $BC$ meet at $R$. Prove that $QR^2 = QA \cdot QB + RA \cdot RD$.

*Solution.*
The right side is the sum of the powers of $Q$ and $R$ with respect to the circle $\omega$ (both points lie outside it, on extensions of chords). With the centre $O$ as origin and $r$ the radius, the power of $Q$ is $|Q|^2 - r^2$, so the claim reads $|Q - R|^2 = |Q|^2 + |R|^2 - 2r^2$, which is equivalent to $Q \cdot R = r^2$. That says $R$ lies on the polar of $Q$, which is Brocard's theorem. L5-GE2 proves this identity again with the Miquel point.

## 1.5 Projective maps

Write the point $(x, y)$ as $[x : y : 1]$, where $[x : y : z] = [\lambda x : \lambda y : \lambda z]$ for $\lambda \neq 0$; the points $[x : y : 0]$ form the line at infinity. A *projective transformation* is a map $[v] \mapsto [Mv]$ with $M$ an invertible $3 \times 3$ matrix. It maps lines to lines and preserves cross-ratios (Chapter 9 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads* develops this).

**Theorem 5.1.**
(1) For any line $\ell$ there is a projective transformation sending $\ell$ to the line at infinity. (2) For any circle $\omega$ and point $P$ inside it, there is a projective transformation sending $\omega$ to a circle



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and $P$ to its centre.

*Proof.* (1) If $\ell$ is $ax + by + cz = 0$, take $M$ with third row $(a, b, c)$ and two further rows making it invertible; then $\ell$ goes to $z = 0$. (2) Similarities are projective, so we may take $\omega$ to be $x^2 + y^2 = z^2$ and $P = [p : 0 : 1]$ with $p \in [0, 1)$. The map $[x : y : z] \mapsto [x - pz : \sqrt{1 - p^2}\, y : z - px]$ multiplies $x^2 + y^2 - z^2$ by $1 - p^2$, so it maps $\omega$ to itself, and it sends $P$ to $[0 : 0 : 1 - p^2]$, the centre. ■

As an illustration, here is a second proof of the quadrilateral harmonic range $(A, D; X, R) = -1$. Send line $QR$ to infinity. Then $AB \parallel CD$ and $AD \parallel BC$, so $ABCD$ becomes a parallelogram and $P$ its centre. Line $QP$ is now the line through the centre parallel to $AB$, so $X$ is the midpoint of $AD$, while $R$ is at infinity; hence $(A, D; X, R) = -1$, and cross-ratio is preserved.

**Common traps.**

- Assuming that the centre of a circle is mapped to the centre of its image.
- Forgetting the special cases of inversion: lines and circles through the centre, and the centre itself.
- Dropping signs in a cross-ratio, or forgetting that a harmonic conjugate can be the point at infinity.
- Using a projective map to deduce a statement about lengths, angles or perpendicularity.

### 1.6 How to spot this in a contest

1. Several circles through one point, or two circles touching at a point: invert there.
2. Tangents from a point together with a secant, or a cyclic quadrilateral with the intersections of opposite sides: think of polars and Brocard.
3. A midpoint together with a parallel line: a harmonic range with a point at infinity.
4. Incircle touch points: the cevians to them concur, so Theorem 3.1(2) applies at once.

Chapter 4 of my book *Beyond the Incenter* treats cross-ratios at greater length, including harmonic bundles, perspectivity and Pascal's theorem.

## 2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** ⋆⋆ Circles of radii 1 and 2 touch externally at $T$, and $\ell$ is the line through their centres. Find the radius of a circle which touches both circles and $\ell$.

**Problem 2** ⋆⋆ Let $A, B, C, D$ be four points, no three collinear and not all on one circle. The angle between two intersecting circles means the acute or right angle between their tangents at a common point. Prove that the angle between circles $ABC$ and $ABD$ equals the angle between circles $CDA$ and $CDB$.



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**Problem 3** ⋆⋆⋆ Let $ABCD$ be a convex quadrilateral with no two sides parallel. Let $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$. Prove that $\angle PRQ = 90^\circ$ if and only if $RP$ bisects angle $ARB$.

**Problem 4** ⋆⋆⋆ Circles $\omega_1$ and $\omega_2$ meet at two points $A$ and $B$. A circle $\Gamma$ touches $\omega_1$ at $P$ and $\omega_2$ at $Q$, where $A, B, P, Q$ are four different points. Prove that $PA \cdot QB = PB \cdot QA$.

**Problem 5** ⋆⋆⋆ Four different circles $\omega_1, \omega_2, \omega_3, \omega_4$ pass through a point $A$. Circles $\omega_1$ and $\omega_3$ touch at $A$, circles $\omega_2$ and $\omega_4$ touch at $A$, and the two common tangents at $A$ are different lines. Circles $\omega_1$ and $\omega_2$ meet again at $P$, circles $\omega_2$ and $\omega_3$ at $Q$, circles $\omega_3$ and $\omega_4$ at $R$, and circles $\omega_4$ and $\omega_1$ at $S$. Prove that

$$AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS.$$

**Problem 6** ⋆⋆⋆⋆ Triangle $ABC$ has circumcentre $O$, circumradius $R$, incentre $I$ and inradius $r$. The incircle touches $BC, CA, AB$ at $D, E, F$, and $H$ is the orthocentre of triangle $DEF$. Prove that

$$\overrightarrow{IH} = -\frac{r}{R}\overrightarrow{IO},$$

so that $H$ lies on line $OI$ (when $O \neq I$), and deduce Euler's formula $OI^2 = R^2 - 2Rr$.

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 2002/03 Problem 2** ⋆⋆
   *Find two similar triangles sharing the angle at A; then compare AP squared with AD times AB.*
2. **BMO2 2017/18 Problem 1** ⋆⋆
   *Invert at B with power BA times BP and ask where M goes.*
3. **BMO2 2013/14 Problem 4** ⋆⋆⋆
   *Find the direction of the tangent at P to circle BCP.*
4. **IMO 2003 Problem 4** ⋆⋆⋆
   *Turn both conditions into statements about DA/DC and BA/BC.*
5. **IMO 2010 Problem 4** ⋆⋆⋆
   *The circle centred at S through C is orthogonal to the circumcircle; it is an Apollonius circle.*
6. **IMO 1996 Problem 2** ⋆⋆⋆
   *Invert at A, or use the inversion distance formula, to turn the angle condition into a length condition.*
7. **USAMO 2015 Problem 2** ⋆⋆⋆
   *Show that AX times AS is constant; inversion at A then swaps X and S.*
8. **IMO 2015 Problem 3** ⋆⋆⋆⋆
   *Invert at H with the negative power that swaps the circumcircle and the nine-point circle.*
9. **USAMO 2017 Problem 3** ⋆⋆⋆⋆
   *Find an inversion centred at S that swaps B and C and also swaps K and M.*



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## 4 Hints

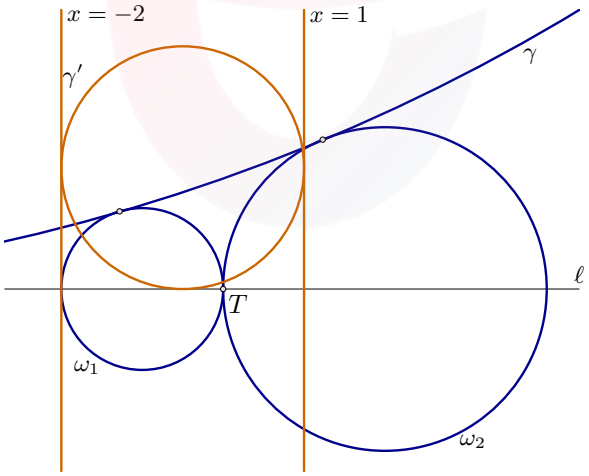
1. Invert at $T$. What do the two circles and $\ell$ become?
2. Three of the four circles pass through $A$.
3. Find a harmonic pencil at $R$ containing the lines $RP$ and $RQ$.
4. Invert at $A$. What can you say about the two tangent lengths from $B'$ to $\Gamma'$?
5. Invert at $A$. What do two circles touching at the centre of inversion become, and what quadrilateral do the four images form?
6. Invert in the incircle. The image of $A$ is a point you know in triangle $DEF$, and the image of the circumcircle is a circle you know.

## 5 Solutions

### Problem 1

*Circles of radii* 1 *and* 2 *touch externally at* $T$*, and* $\ell$ *is the line through their centres. Find the radius of a circle which touches both circles and* $\ell$*.*

Use coordinates with $T$ at the origin and $\ell$ the $x$-axis, so the centres are $(-1,0)$ and $(2,0)$. Invert at $T$ with $r^2 = 4$. The first circle passes through $T$ and $(-2,0)$, which is fixed, so it becomes the line $x = -2$ (perpendicular to $\ell$, by symmetry). The second passes through $T$ and $(4,0)$, whose image is $(1,0)$, so it becomes the line $x = 1$. The line $\ell$ passes through $T$ and is fixed.




A circle $\gamma$ touching both circles and $\ell$ cannot pass through $T$: it would have to touch both circles at $T$, so its tangent at $T$ would be the common tangent, which is perpendicular to $\ell$, and then $\gamma$ would cross $\ell$ at $T$ instead of touching it. So $\gamma'$ is a circle touching $x = -2$, $x = 1$ and $\ell$ (Theorem 1.3). It therefore has radius $\frac{3}{2}$ and centre $(-\frac{1}{2}, \pm\frac{3}{2})$; by symmetry take the plus sign.

The power of $T$ with respect to $\gamma'$ is $k = \frac{1}{4} + \frac{9}{4} - \frac{9}{4} = \frac{1}{4}$. By the proof of Theorem 1.2(3), $\gamma$ is the image of $\gamma'$ under the homothety at $T$ with ratio $r^2/k = 16$, so its radius is $16 \cdot \frac{3}{2} = 24$ and its centre is $(-8, 24)$. As a check, the distances from $(-8,24)$ to $(-1,0)$ and $(2,0)$ are $25 = 24 + 1$ and $26 = 24 + 2$. There are two such circles, one on each side of $\ell$, both of radius 24.

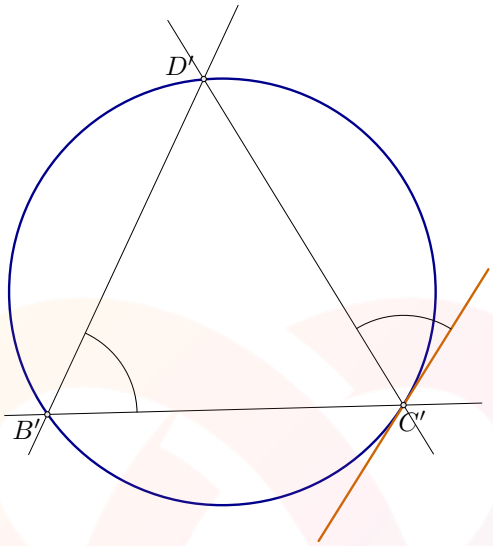


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**Answer:** 24

### Problem 2

*Let $A, B, C, D$ be four points, no three collinear and not all on one circle. The angle between two intersecting circles means the acute or right angle between their tangents at a common point. Prove that the angle between circles $ABC$ and $ABD$ equals the angle between circles $CDA$ and $CDB$.*




Invert at $A$ with any radius. Circles $ABC$, $ABD$ and $CDA$ pass through $A$, so they become the lines $B'C'$, $B'D'$ and $C'D'$. Circle $CDB$ does not pass through $A$, since the points are not concyclic, so it becomes the circle $C'D'B'$. The points $B', C', D'$ are not collinear, since otherwise circle $BCD$ would pass through $A$.

By Theorem 1.3, the angle between circles $ABC$ and $ABD$ at their common point $B$ equals the angle between lines $B'C'$ and $B'D'$, which is the acute or right angle determined by $\angle C'B'D'$. The angle between circles $CDA$ and $CDB$ at their common point $C$ equals the angle at $C'$ between the line $C'D'$ and the circle $C'D'B'$, that is, between the chord $C'D'$ and the tangent at $C'$. By the alternate segment theorem this angle is $\angle C'B'D'$ or its supplement, so it determines the same acute or right angle. Hence the two angles are equal.

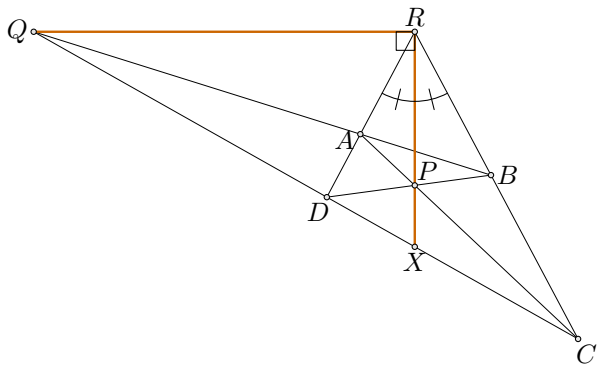
### Problem 3

*Let $ABCD$ be a convex quadrilateral with no two sides parallel. Let $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$. Prove that $\angle PRQ = 90^\circ$ if and only if $RP$ bisects angle $ARB$.*

The relabelling $A \leftrightarrow D$, $B \leftrightarrow C$ leaves $P$, $Q$, $R$ unchanged, and angle $ARB$ is the same as angle $DRC$, since by convexity $A, D$ lie on one ray from $R$ and $B, C$ on another. So we may assume $A$ lies between $R$ and $D$, and then, by convexity, $B$ lies between $R$ and $C$, and the quadrilateral lies inside triangle $RDC$.



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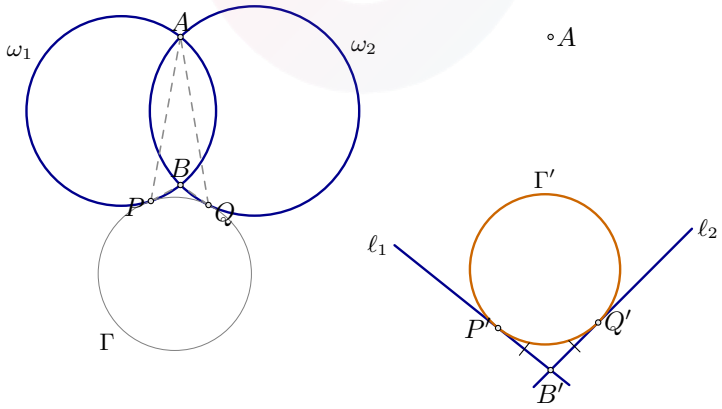

In triangle $RDC$ the points $A$ and $B$ lie on sides $RD$ and $RC$, and the cevians $DB$ and $CA$ meet at $P$. Let $RP$ meet $DC$ at $X$; since $P$ is inside the triangle, $X$ lies on segment $DC$. The line $AB$ meets $DC$ at $Q$. By Theorem 3.1(2), $(D, C; X, Q) = -1$. As $Q$ is a finite point, $X$ is not the midpoint of $DC$.

Suppose $\angle PRQ = 90°$. Then $\angle XRQ = 90°$ and $(D, C; X, Q) = -1$, so by Theorem 3.1(1) the lines $RX$ and $RQ$ bisect the angles between $RD$ and $RC$. Since $X$ lies on segment $DC$, $RX$ is the internal bisector of angle $DRC$, that is, $RP$ bisects angle $ARB$.

Conversely, suppose $RP$ bisects angle $ARB = DRC$, so $RX$ is the internal bisector. Then $RD/RC = XD/XC \neq 1$, so the external bisector at $R$ is not parallel to $DC$ and meets it at a point $Q_1$ with $(D, C; X, Q_1) = -1$ by Theorem 3.1(1). The harmonic conjugate of $X$ with respect to $D, C$ is unique, so $Q_1 = Q$. Hence $RQ$ is the external bisector, which is perpendicular to the internal bisector $RP$, and $\angle PRQ = 90°$.

### Problem 4

*Circles $\omega_1$ and $\omega_2$ meet at two points $A$ and $B$. A circle $\Gamma$ touches $\omega_1$ at $P$ and $\omega_2$ at $Q$, where $A, B, P, Q$ are four different points. Prove that $PA \cdot QB = PB \cdot QA$.*




Invert at $A$ with radius 1. Since $\omega_1$ and $\omega_2$ pass through $A$ and $B$, their images are two different lines $\ell_1, \ell_2$ through $B'$. The circle $\Gamma$ does not pass through $A$: the only common point of $\Gamma$ and $\omega_1$ is $P$, and $P \neq A$. So $\Gamma'$ is a circle, and by Theorem 1.3 it touches $\ell_1$ at $P'$ and $\ell_2$ at $Q'$. Similarly $B$ is not on $\Gamma$ (as $B \neq P$), so $B'$ is not on $\Gamma'$.

Hence $B'P'$ and $B'Q'$ are the two tangent segments from $B'$ to $\Gamma'$, and $B'P' = B'Q'$. By Theorem 1.1,



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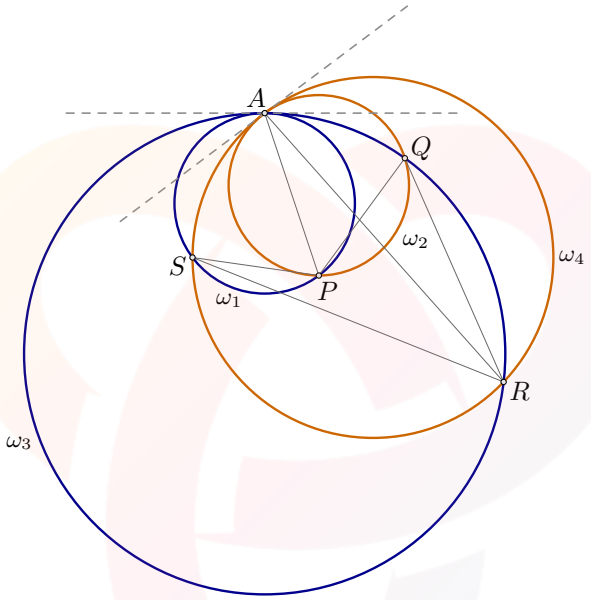
$$B'P' = \frac{BP}{AB \cdot AP}, \qquad B'Q' = \frac{BQ}{AB \cdot AQ}.$$

Equating and multiplying by $AB \cdot AP \cdot AQ$ gives $BP \cdot AQ = BQ \cdot AP$, as required. In other words, $P$ and $Q$ lie on the same Apollonius circle of $A$ and $B$.

### Problem 5

*Four different circles $\omega_1, \omega_2, \omega_3, \omega_4$ pass through a point $A$. Circles $\omega_1$ and $\omega_3$ touch at $A$, circles $\omega_2$ and $\omega_4$ touch at $A$, and the two common tangents at $A$ are different lines. Circles $\omega_1$ and $\omega_2$ meet again at $P$, circles $\omega_2$ and $\omega_3$ at $Q$, circles $\omega_3$ and $\omega_4$ at $R$, and circles $\omega_4$ and $\omega_1$ at $S$. Prove that*

$$AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS.$$




First the points exist. The circles $\omega_1$ and $\omega_2$ have different tangents at $A$, so they do not touch there and meet at a second point $P \neq A$. Similarly $Q$, $R$ and $S$ exist and differ from $A$.

Invert at $A$ with radius 1. By Theorem 1.2 each $\omega_i$ is mapped to a line $\ell_i$ not through $A$, parallel to the tangent of $\omega_i$ at $A$. Two circles that touch at $A$ have no common point other than $A$, which goes to $\infty$, so $\ell_1$ and $\ell_3$ have no common point: they are different parallel lines. Similarly $\ell_2 \parallel \ell_4$. As the tangents of $\omega_1$ and $\omega_2$ at $A$ are different, $\ell_1$ and $\ell_2$ are not parallel. The second intersection $P$ of $\omega_1$ and $\omega_2$ goes to the only common point of $\ell_1$ and $\ell_2$, so $P' = \ell_1 \cap \ell_2$, and in the same way $Q' = \ell_2 \cap \ell_3$, $R' = \ell_3 \cap \ell_4$, $S' = \ell_4 \cap \ell_1$.

Hence $P'Q'R'S'$ is a parallelogram whose sides $P'Q'$, $Q'R'$, $R'S'$, $S'P'$ lie on $\ell_2$, $\ell_3$, $\ell_4$, $\ell_1$; its four vertices are different, since any two of them lie on $\ell_1$ and $\ell_3$ respectively, or on $\ell_2$ and $\ell_4$, and these pairs of lines are disjoint. Opposite sides of a parallelogram are equal, so $P'Q' = R'S'$ and $Q'R' = S'P'$.

Now Theorem 1.1 (which also holds when $A$ and the two points are collinear) gives $P'Q' = \frac{PQ}{AP \cdot AQ}$, and similarly for the other three sides. The two equalities become

$$\frac{PQ}{AP \cdot AQ} = \frac{RS}{AR \cdot AS'}, \qquad \frac{QR}{AQ \cdot AR} = \frac{SP}{AS \cdot AP}.$$

Dividing the first by the second, $AQ$ and $AS$ cancel and we get



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$$\frac{PQ \cdot AR}{QR \cdot AP} = \frac{RS \cdot AP}{SP \cdot AR},$$

that is, $AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS$.

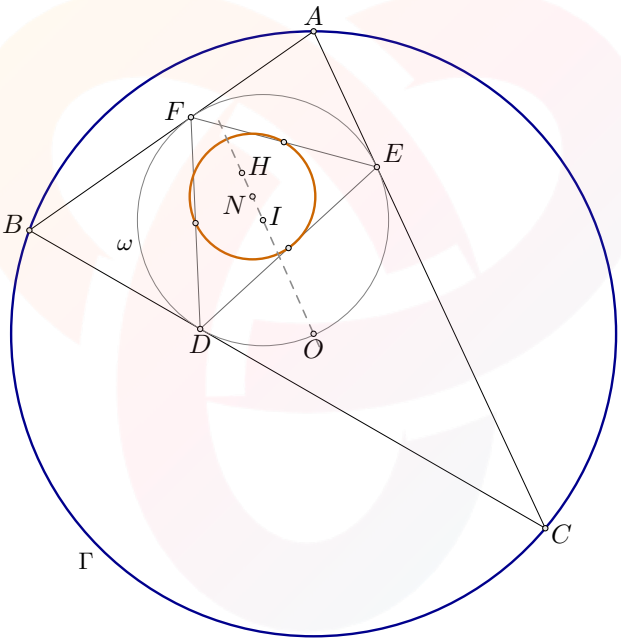
Each displayed equality is a result on its own, for example $PQ \cdot AR \cdot AS = RS \cdot AP \cdot AQ$, and the problem needs both. In problems of this kind, write down every simple fact about the inverted figure (here: opposite sides parallel and equal) and translate each one back with Theorem 1.1.

### Problem 6

*Triangle $ABC$ has circumcentre $O$, circumradius $R$, incentre $I$ and inradius $r$. The incircle touches $BC, CA, AB$ at $D, E, F$, and $H$ is the orthocentre of triangle $DEF$. Prove that*

$$\overrightarrow{IH} = -\frac{r}{R}\,\overrightarrow{IO},$$

*so that $H$ lies on line $OI$ (when $O \neq I$), and deduce Euler's formula $OI^2 = R^2 - 2Rr$.*




Let $\omega$ be the incircle and invert in $\omega$. The point $A$ lies outside $\omega$ with tangents $AE$ and $AF$, so its inverse is the point where $AI$ meets the chord of contact $EF$ (Section 4). As $AE = AF$ and $IE = IF$, line $AI$ is the perpendicular bisector of $EF$, so this point is the midpoint of $EF$. Similarly $B$ and $C$ go to the midpoints of $FD$ and $DE$. The circumcircle $\Gamma$ does not pass through $I$ (which is inside it), so by Theorem 1.2 its image $\Gamma'$ is the circle through the three midpoints of the sides of $DEF$, that is, the nine-point circle of $DEF$.

We use the standard facts about the nine-point circle of a triangle: its radius is half the circumradius, and its centre is the midpoint of the segment joining the circumcentre to the orthocentre. The circumcircle of $DEF$ is $\omega$, with centre $I$ and radius $r$, so $\Gamma'$ has radius $r/2$ and centre $N$, the midpoint of $IH$.

Now compute $\Gamma'$ from the proof of Theorem 1.2(3). The power of $I$ with respect to $\Gamma$ is $k = OI^2 - R^2$, which is negative as $I$ is inside $\Gamma$. So $\Gamma'$ is the image of $\Gamma$ under the homothety with centre $I$ and ratio $r^2/k$. Comparing radii, $R \cdot r^2/|k| = r/2$, so $|k| = 2Rr$, that is, $R^2 - OI^2 = 2Rr$, which is Euler's formula. Comparing centres,



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$$\overrightarrow{IN} = \frac{r^2}{k}\overrightarrow{IO} = -\frac{r^2}{2Rr}\overrightarrow{IO} = -\frac{r}{2R}\overrightarrow{IO}.$$

Since $N$ is the midpoint of $IH$, $\overrightarrow{IH} = 2\overrightarrow{IN} = -\frac{r}{R}\overrightarrow{IO}$. If $O \neq I$ this puts $H$ on line $OI$, on the other side of $I$ from $O$; if $O = I$ (the equilateral case) it says $H = I$, which is also true.